lassa — Extracted Transmission Models

Overview

A total of 86 transmission models were extracted from 43 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (80.2%), with Vector/Animal to human (18.6%) representing the most commonly modelled transmission route.

Summary

Metric	Value
Models extracted	86
Articles considered	43
Deterministic models	62 (72.1%)
Stochastic models	22 (25.6%)
Models with available code	17 (19.8%)

Figure 1: Model Overview

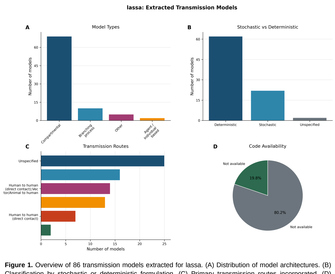

Figure 1. Overview of 86 transmission models extracted for lassa. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

Model Formulation

Deterministic formulations were employed in 72.1% of models, while stochastic approaches accounted for 25.6%. Empirical data were incorporated in 48.8% of the extracted models, with 53.5% classified as purely theoretical.

Figure 2: Interventions and Assumptions

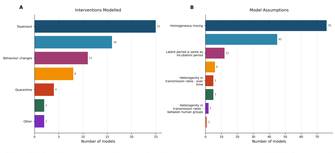

Figure 2. Interventions and assumptions in extracted lassa transmission models (n=86). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

Interventions and Assumptions

The most frequently modelled interventions were Treatment (29.1%), Vaccination (18.6%), Behaviour changes (12.8%). Common modelling assumptions included Homogeneous mixing (88.4%), Latent period is same as incubation period (14.0%), Age dependent susceptibility (7.0%).

Figure 3: Programming Languages

Figure 3. Programming languages used for model implementation among 47 models with specified language.

Figure 4: Model Type by Formulation

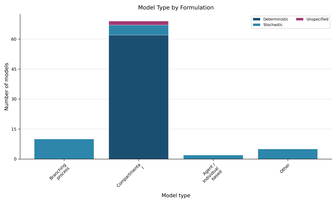

Figure 4. Distribution of deterministic and stochastic formulations across model types (n=86).

Reproducibility

Source code was publicly available for 19.8% of the extracted models.

Tables

Table 1: Model Types

Model Type	Count	Proportion
Compartmental	69	80.2%
Branching process	10	11.6%
Other	5	5.8%
Agent / Individual based	2	2.3%

Distribution of model architectures across 86 extracted transmission models.

Table 2: Model Formulation

Formulation	Count	Proportion
Deterministic	62	72.1%
Stochastic	22	25.6%
Unspecified	2	2.3%

Classification of extracted models by stochastic versus deterministic formulation (n=86).

Table 3: Transmission Routes

Transmission Route	Count	Proportion
Unspecified	25	29.1%
Vector/Animal to human	16	18.6%
Human to human (direct contact);Vector/Animal to human	14	16.3%
Vector/Animal to human;Human to human (direct contact)	13	15.1%
Human to human (direct contact)	7	8.1%
Vector/Animal to human;Human to human (direct non-sexual contact)	2	2.3%
Human to human (direct contact);Vector/Animal to human;Airborne or close contact	2	2.3%
Vector/Animal to human;Sexual	2	2.3%
Human to human (direct contact);Vector/Animal to human;Airborne or close contact;Unspecified	1	1.2%
Vector/Animal to human;Sexual;Airborne or close contact	1	1.2%
Human to human (direct non-sexual contact);Vector/Animal to human	1	1.2%
Human to human (direct contact);Airborne or close contact	1	1.2%
Sexual;Vector/Animal to human	1	1.2%

Primary transmission routes incorporated in extracted models (n=86).

Table 4: Interventions Modelled

Intervention Type	Count	Proportion
Treatment	25	29.1%
Vaccination	16	18.6%
Behaviour changes	11	12.8%
Vector/Animal control	8	9.3%
Quarantine	4	4.7%
Hospitals	2	2.3%
Other	2	2.3%

Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=86).

Table 5: Model Assumptions

Assumption	Count	Proportion
Homogeneous mixing	76	88.4%
Other	45	52.3%
Latent period is same as incubation period	12	14.0%
Age dependent susceptibility	6	7.0%
Heterogeneity in transmission rates - over time	5	5.8%
Heterogeneity in transmission rates - between groups	5	5.8%
Heterogeneity in transmission rates - between human groups	2	2.3%
Heterogeneity in transmission rates - between human and vector	1	1.2%

Common modelling assumptions. Categories are not mutually exclusive (n=86).

Table 6: Code Availability

Code Available	Count	Proportion
Yes	17	19.8%
No	69	80.2%

Availability of publicly accessible source code for extracted models (n=86).

Table 7: Programming Languages

Programming Language	Count	Proportion
Unspecified	39	45.3%
R	20	23.3%
Matlab	20	23.3%
Other	5	5.8%
C++	1	1.2%
Python	1	1.2%

Programming languages used for model implementation (n=86).

Table 8: Empirical Data Usage

Empirical Data Used	Count	Proportion
Yes	42	48.8%
No/Unspecified	44	51.2%

Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 86 extracted transmission models for lassa is publicly available.

Table 9: Sample of Extracted Models

Article ID	Model Type	Compartmental Structure	Formulation	Transmission Route
PMID_25569707	Branching process	Not compartmental	Stochastic	Human to human (direct contact);Vector/A
DOI_7bc8ecc4cac7	Compartmental	Other compartmental	Deterministic	Human to human (direct contact);Vector/A
PMID_25229705	Compartmental	Other compartmental	Deterministic	Unspecified
PMID_25229705	Compartmental	Other compartmental	Deterministic	Unspecified
PMID_33657095	Compartmental	SIR	Deterministic	Vector/Animal to human
DOI_9f38d3d860db	Compartmental	SEIR	Deterministic	Human to human (direct contact);Vector/A
PMID_31007139	Agent / Individual based	Not compartmental	Stochastic	Unspecified
PMID_27588425	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_27588425	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_27588425	Other	Not compartmental	Stochastic	Vector/Animal to human

Sample of 10 records from 86 extracted lassa transmission models.

Data Availability

The complete dataset of extracted transmission models for lassa (n=86) has been made publicly available to support future modelling efforts and outbreak preparedness.